

A second-order intervention

Amanda Corris¹ · Anthony Chemero¹

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Most of Shaun Gallagher’s *Enactivist Interventions* into philosophical issues about the mind are quite effective, but there are a few that could be improved. In what follows, we attempt to make two of Gallagher’s arguments more convincing. Our focus will be on Gallagher’s use of Francisco Varela’s “threefold distinction in temporal and dynamical registers” (Gallagher 2017, 8; Varela 1999). Gallagher uses this threefold distinction to address issues concerning what he calls the “causal-constitution fallacy” and issues concerning neuroscientific findings and free will. Gallagher is less convincing than he could be when he addresses these issues, because although he invokes Varela’s threefold distinction, he does not also provide a detailed story about how these “temporal and dynamical registers” relate to one another. We will provide a counter-intuitive, but empirically well-supported story about how the registers relate to one another, and, in so doing, improve upon Gallagher’s interventions.

We begin with Varela’s threefold distinction. What Varela calls the *elementary* timescale consists of neurophysiological measures, such as the “rhythms of bursting interneurons” (9). On this timescale, which varies from 10 to 500 or so ms, basic neuronal and cellular processes occur. The *integrative* timescale, varying from half a second to 3 s, corresponds to neuronal processes integrated with cell assemblies, thus encompassing “the level of a fully constituted cognitive operation” (9). The integrative timescale is used to describe basic actions such as reaching and grasping. The *narrative* scale encompasses “complex actions and cognitive processes that may involve recollection, planning, intention formation, and so on” (9). Processes occurring on the narrative scale do so after 3 s, and hypothetically could last much longer. In this way, cognitive processes are distributed across different timescales and are “constituted in a temporally integrated dynamical system” (10). Each

✉ Anthony Chemero
tonychemero@gmail.com

¹ University of Cincinnati, Cincinnati, USA

timescale is integrated into the next, allowing for a ‘scaling up’ of explanatory accounts of mental processes. While these three timescales make up the framework relevant to an enactive account of cognition, Gallagher notes that further timescales (such as developmental and evolutionary timescales) could play explanatory roles as well. One feature of Varela’s distinction that Gallagher does not explicitly note, but that will be important to us below, is that in cases that are relevant to cognition these temporal scales have corresponding spatial scales. Fast elementary timescale events are also small, e.g., neuron firings and other cellular activities; events at the integrative timescale are larger, like whole brain activity, reflexes, and eye moments; events at the narrative timescale are larger still, e.g., planning a party, which would include making lists and walking over to check what’s left in the refrigerator.

Gallagher first uses the threefold distinction to counter the argument that enactivist approaches to cognition commit what he calls the causal-constitution (C–C) fallacy.¹ Enactivists like Gallagher typically believe that artifacts that are closely involved in our thinking are sometimes constitutive parts of that thinking. Satoshi doesn’t need to use his brain to remember phone numbers because they are stored in his phone. It seems natural to say in cases like this that Satoshi’s cognitive act of remembering a phone number is accomplished by interacting with his phone, so his interaction with the phone and the phone itself are parts of the remembering. Those who say this is fallacious say it makes an “unjustifiable inference from *causal* dependence (where bodily and environmental factors play a causal role in support of cognitive processes) to *constitutive* dependence (where the claim is that such factors actually are part of the cognitive processes)” (7). Causal relations, the claim is, do not entail constitutive relations and things like phones are not the right sort of things to be parts of cognitive processes; only things like brains can be parts of cognitive processes. In this case, it is Satoshi’s neural representation of the phone, not the phone itself, that is part of the cognitive process. As Gallagher puts the objection, an emotion, for example, might be “causally influenced by bodily or environmental factors, but what constitutes an emotion is just the mental event that is instantiated by the brain” (8). Losing a favorite lizard might be the cause of one’s sadness, but that loss is not a constitutive part of the sadness. More generally, suggesting that environmental causal factors and other things of the wrong kind count as part of the cognitive process is committing the C–C fallacy. Gallagher counters this by pointing out that it depends upon a synchronic relation between cause and constitution, in that “[i]f x constitutes (or composes) y, then x and y exist at the same place at the same time and they share the same material parts” (8). However, this view of cognition is problematic, as mental activities are processes, and a synchronic understanding of constitution cannot account for processes that take time, and because they might be constituted by changing parts over time.

Gallagher makes use of Varela’s threefold distinction in order to avoid the C–C fallacy. According to the charge of the C–C fallacy, enactivists confuse causal and constitutive relations. The temporal framework, however, relies upon a notion of

¹ Most people call this the coupling-constitution fallacy, following Adams and Aizawa (2001).

diachronic constitution, rather than synchronic. Mental processes are “distributed across different factors/levels (neural, behavioral, environmental), and across different timescales”, and thus are “constituted in a temporally integrated dynamical system” (9–10). Describing mental processes as being constituted diachronically, across explanatorily distinct but integrated timescales, allows for what Gallagher calls a “more nuanced distributed-state materialism” (9), where constitutive factors “may very well be in complex, reciprocal causal relations with each other, but just these reciprocal causal relations make the mental process what it is” (10).

Describing cognitive processes as being constituted in this temporally-integrated dynamic system is also helpful in explaining more precisely the relationship between cognition and action. Gallagher makes reference to the elementary timescale in discussing conclusions about free will stemming from the infamous Libet experiments, ultimately arguing that free will is “best understood on a narrative timescale” (25). By identifying the appropriate timescale for particular cognitive processes, enactivists can provide explanations for those processes that appropriately describe the roles the brain, body, and environment play in cognition.

The current debate over free will in many quarters is structured by a series of experiments performed by Benjamin Libet and his colleagues. (Libet 1985 is a good, early overview). Libet’s experiments focus on the measurement of motor control processes and equate those with agency. In one experiment, subjects are told to flick their wrists whenever they wish to, reporting when they made the conscious decision to do so, while their brain activity is monitored. Before the motor action of flicking one’s wrist, there are two relevant neurological events that researchers were able to measure: several hundred ms of brain activity known as the readiness potential, followed by 50 ms of activity in the motor nerves descending from the motor cortex to the wrist. The experiment showed that there was on average a period of 350 ms before subjects were conscious of their decision to move when their brains already initiated processes involved in motor action. Libet concluded from this experiment that even before subjects reported making the conscious decision to move, the brain had already subconsciously begun the processes involved in moving—it “seemingly decides and then initiates its decisions in a non-conscious fashion within a tight elementary timescale” (139). It seems possible to conclude that because the brain will have already initiated the motor process before a subject becomes consciously aware of the decision to move, free actions, and thus free will, are in some sense illusory. The brain, as Gallagher suggests we might be tempted to think, “inventively tricks us into thinking that we consciously decide matters” (139). While Libet did conclude that there remains room for free will in the approximately 150 ms between the conscious experience of intending to move and the actual movement—what Libet calls the ‘conscious veto’ (Libet 1999, 52)—the suggestion that our brains essentially trick us into believing we are controlling our motor actions has generated significant debate.

Gallagher stresses, we think correctly, that this is the wrong approach. Motor control processes, as neurophysiological processes, occur in the elementary timescale. We generally have no conscious awareness of them. Gallagher notes that “in the normal situation we do not require a second-order consciousness of

bodily movement; we do not have to be reflectively conscious of the onset of the action or the course of the movement as we execute it” (141). Free actions are those which we consciously experience having some control or influence over. But associating free will with motor control becomes problematic, since, as Gallagher points out, most motor control is not typically performed consciously. Questions about free will are not questions about motor control. The mistake, Gallagher holds, is looking for free will in the domain of fast, small neurophysiological events—“free will cannot be squeezed into the elementary timescale of 150–350 ms; free will is a longer term phenomenon” (143). Instead, Gallagher suggests we look to the integrative and narrative timescales to describe actions relevant to the question of free will. Doing so reframes the question in such a way that makes room for agency. An intentional action will necessarily involve neurophysiological processes occurring in the elementary timescale but will also necessarily involve processes occurring in the integrative and narrative timescales. Questions about free will focus on intentional actions—“the kinds of processes associated with free actions are not made at the spur of the moment—they are not elementary or momentary and cannot fit within the thin phenomenology of the ms between [readiness potential] and movement” (143). To make sense of free will, explanations of free, intentional actions must take into account processes on a longer timescale, not merely the fast neurophysiological processes associated with motor control. These actions can be “described at the most appropriate pragmatic level of description” (143) by making reference to the integrative and narrative timescales.

To illustrate this point, Gallagher discusses a scenario that “reflects the distinction between fast movement under automatic control on the elementary timescale, and slower voluntary action on integrative and narrative timescales” (143–144). A movement in the grass near him elicits a reflexive response approximately 200 ms after the stimulus. This action is not subject to conscious control: his movement “can be explained purely in terms of non-conscious perceptual processes, neuronal firing, and muscles contracting, ... etc.” (144). Between one and 5 s later, responses at the integrative and narrative timescales occur—Gallagher recognizes that the stimulus was the movement of a harmless lizard, which prompts him to consider catching it for his collection,² and it is only a half second later, more than 5 s after the stimulus, that he *voluntarily* reaches for the lizard. The free action of interest is the identification of and decision to catch the lizard, which is a process that occurs over the course of a timeframe that extends far beyond the ten to 500 ms timeframe characterized by the elementary timescale. Gallagher stresses that the “proper level of description relevant to free will” will not make reference solely to motor actions but must incorporate the more complex cognitive processes associated with decision making as well. These processes are extended through the integrative and narrative timescales. Reframing the question of free will in terms of these longer timescales allows for an alternative approach to the problems raised by the Libet experiments—one that does not relegate discussion of “free” action to motor control. The temporal-dynamic register

² Gallagher’s lizard collection? We cannot say enough how much we love this, and fervently hope that Gallagher really has a lizard collection. If not, Gallagher could consider an alternative career as a Dada poet. We must admit, though, that we were less impressed with “Libetarian”.

framework, then, is a useful tool for reconsidering classic views regarding free will and cognitive science.

Gallagher's interventions in these two cases rely on the same point: small and fast events constitute large and slow events, and we can't read features of the latter off of features of the former. In the case of the C–C fallacy, at short temporal and spatial scales, we might be able to locate merely neural events that constitute an elementary sliver of a cognitive process merely causally interacting with embodied and environmental elementary slivers, but this is the wrong spatial and temporal scale at which to understand cognition. In the case of the neuroscience of free will, while it might be the case that neural events at the elementary scale seem confounding, that time scale is not the appropriate one for considering the experiences involved in the conscious control of action, which can only occur at slower time scales. We think these are the right points to be making. At the same time, they both beg the same question: what exactly is the relationship between the fast and small events and the slow and large events? In both cases, the relationship must be such that events at the elementary scale are constitutive of events at integrative and narrative scales, but are also (in part at least) controlled by events at the integrative and narrative scales. This is easiest to see in the case of the challenge to free will posed by experiments by Libet and colleagues. How are temporally extended events at the integrative and narrative scale able to influence the elementary-scale events that they are constituted by? How does recognizing an attractive lizard lead to the neural and muscular events involved in trying to catch it? To use Gallagher's words from above, both voluntary movements and reflexes include "non-conscious perceptual processes, neuronal firing, and muscles contracting, ... etc." at elementary scales. But in the former case, but not the latter, they are causally influenced by the integrative and narrative scale events which they partly constitute. Otherwise, the integrative and narrative scales are epiphenomenal. Gallagher's response to the C–C fallacy is in the same boat. If actual bodily behavior is generated by events at the elementary scale, what role do the constituted brain-body-environment processes at the integrative and narrative scale play? Many philosophers would rule macro-to-micro causation an impossibility. [This seems to be a large part of the motivation for neo-mechanistic philosophy of science (Bechtel and Richardson 1993/2010; Craver and Bechtel 2007)]. Gallagher's intervention on issues concerning free will and the brain requires an account of macro-to-micro causal influence. Our aim in the rest of this commentary is to provide one.

Because many philosophers would dismiss the idea of macro-to-micro causal influence, we will begin with a homey example: the whirlpool that forms when one flushes a toilet. When you flush the toilet, you release a large amount of potential energy, from the elevated water in the tank, into the comparatively low energy bowl of water. Once this happens, a whirlpool forms in the bowl until the potential energy from the water in the tank has dissipated, over the course of a few seconds. The individual water molecules whose activities constitute the whirlpool, which had previously been moving about randomly (i.e., in Brownian motion), all begin to behave in statistically highly unlikely ways: they collectively move in a helical pattern, and they do so until the potential energy from the water in the tank has dissipated. The key point is that the water molecules, at the elementary scale, are constrained to behave in very unlikely ways by the larger, slower whirlpool that

they collectively constitute. Complexity theorists describe systems that engage in this macro-to-micro causal influence as ‘synergies’. A synergy is a group of elements temporarily constrained to act as a unit (Haken 1977; Kelso 1995), in which the comparatively large-scale, comparatively slow processes partly determine the behavior of comparatively small-scale, comparatively fast processes that constitute them. A second key point is that it is thoroughly established that synergies are ubiquitous in nature: along with physical phenomena like whirlpools, cells, organs, and organisms are all synergies. There is growing empirical evidence that there are also human-tool synergies and interpersonal synergies (Dotov et al. 2010; Riley et al. 2011). In each of these cases, the macro-scaled process as a whole constrains the activity of the micro-scaled processes that constitute it. Notice that ‘constrain’ does not mean precisely determine. Consider the diurnal cycle: your day-long cycle makes it likely that you will be asleep at particular times and hungry at other times, but you sometimes still want a late-night snack. A third point here is that if the next few paragraphs seem counterintuitive, we encourage readers to remember the utterly mundane whirlpool in the toilet.³

Although there is a growing literature on synergies in the cognitive and neural sciences,⁴ the work most relevant to current concerns was done by Guy van Orden and his colleagues. In one series of experiments, Van Orden et al. (2003) had participants engage in just over one-thousand trials(!)⁵ of a temporal estimation task: try to press the button exactly one second after you hear the signal to start. The main data were the errors: how much each estimate differed from exactly one second. The key for current purposes is that the experiment encompassed narrative, integrative, and elementary scales. The twenty or so minutes that the participant spent as a time estimator were constituted by a thousand button presses at the integrative scale which were in turn constituted by countless neural, hormonal, muscular, technological, etc. events at the elementary scale. Van Orden et al. found that the time series of estimates had a self-similar or fractal structure. That is, the pattern of variability in the entire one-thousand estimate experiment was similar to that found when the series of measurements was broken into five-hundred trial chunks, which was similar to the pattern of variability when it was broken into two-hundred-fifty trial chunks, which was similar to the pattern when it was broken into one-hundred-twenty-five trial chunks, and so on. This is evidence that the experiment as a whole was constraining the chunks of trials, and through them,

³ A fourth point, which we can’t discuss here, is that synergies are an explanation of the existence of ordered processes, like those that constitute living things, in a universe ruled by the second law of thermodynamics. Synergies occur when systems are temporarily far from thermodynamic equilibrium, i.e., energy is not evenly distributed. The whirlpool in the toilet occurs because of the potential energy released from the tank and ends when the energy is dissipated; living things temporarily maintain their processes by constantly concentrating energy in themselves (by eating, photosynthesizing, etc.). Deacon (2011) explains this more accessibly than is typical.

⁴ See Anderson et al. (2012) for a slightly dated review.

⁵ 1024 for those who care. The data analysis requires three orders of magnitude, and the math is simplified by having the number of trials be a power of 2. See van Orden et al. (2003) for details.

the individual trials that constituted it.⁶ That is, the participant-with-button-and-software that completed the experiment was a synergy temporarily constituted by smaller scale entities and events constrained to work together by being part of that synergy.

Van Orden et al. (2012) realized that synergies in behavior contradicted the brain's widely assumed status as the executive that controls behavior. Even though dualism is no longer respectable in the cognitive and neural sciences, the mainstream understanding of the brain is that it does all the work that immaterial minds were supposed to do: reasoning, planning, emoting, and generally being in charge of the body.⁷ If a behaving human is a synergy at the narrative scale, then the neural processes that partly compose that synergy are constrained by the behavior; they do not control it. Van Orden et al. call it the "blue-collar brain". Rather than being the executive in charge of the body, the brain does what is required by the activities of the embodied person. The way in which the brain is controlled by behavior is not in any way like the ways that we have mythologized the brain's supposed control of the body, which is supposed to be akin to a computer sending precise motor commands to muscle groups. Instead, a person's ongoing activity constrains the brain's activity the way a whirlpool controls that of individual water molecules or the way the diurnal cycle controls hunger and sleepiness. In a whirlpool, most of the water molecules move in a more or less helical trajectory. Similarly, while you are walking to your office your brain's activities will be such that they reliably enable your gait, upright posture, awareness of obstacles, etc., but the activity of walking to your office will not constrain your brain's activity in full detail, down to the neuron.

We admit that the blue-collar brain—the idea that elementary-scale events in the brain do not control narrative-scale behavior, and instead, the fast and small processes in the nervous system are constrained by the slow and big activities of the embodied person—is a dramatic departure from what most cognitive and neural scientists believe about the role of the brain in behavior. We recommend it to Gallagher for several reasons. First, there is ample and growing evidence that synergies are ubiquitous in biological systems (see Kelso 2008; Deacon 2011; West 2016). Second, there is also ample and growing evidence that synergies are ubiquitous in large-scale (i.e., integrative) neural activity (Anderson et al. 2012; Anderson 2014) and in human individual and interpersonal behavior (Abney et al. 2014; Paxton and Dale Paxton and Dale 2013; Nalepka et al. 2017; Chemero 2016). This implies that the fast and small activities of neurons are constrained by slower and bigger integrative activities. Third, and most importantly, Gallagher's interventions build upon a helpful set of distinctions between temporal and spatial scales, but there is also a need for a story about the interactions between these

⁶ That this self-similarity in the time series is evidence of a synergy was controversial for a few years. See Wagenmakers et al. (2004) and van Orden et al. (2005). Ihlen and Vereijken (2010) ends the controversy fairly definitely.

⁷ One of the reasons van Orden et al. think that the idea of the brain causing the body to move is a myth is that there is no way that brain activity could literally cause bodily movements: the amount of energy in the firing of neurons is simply insufficient to lift a 10 kg leg off the ground.

scales. Otherwise, Gallagher has no way to account for how the narrative processes actually make things happen in the real-time activities of an embodied person. Gallagher's interventions into the C–C fallacy and the debates over free will require the blue-collar brain, or something very similar.

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